Module 8: Science and Society

Outcomes

A student:

› analyses and evaluates primary and secondary data and information INS11/12-5
› solves scientific problems using primary and secondary data, critical thinking skills and scientific processes INS11/12-6
› communicates scientific understanding using suitable language and terminology for a specific audience or purpose INS11/12-7
› evaluates the implications of ethical, social, economic and political influences on science INS12-15

Related Life Skills outcomes: SCLS6-5, SCLS6-6, SCLS6-7, SCLS6-14, SCLS6-15

Content Focus

Those who pursue the study of science have created processes, tools and products that challenge and influence society and some of its belief systems, ethics and societal norms. In response, society debates and regulates science in order to prevent harmful developments and unacceptable outcomes, and to allow for new and beneficial products, processes and ideas. Science also can be affected by society, as well as governments, industry, economic interests and cultural perspectives.

Students explore the impacts of ethical, social, economic and political influences on science and its research.

Working Scientifically

In this module, students focus on analysing and evaluating primary and secondary data to solve problems and communicate scientific understanding about the position and application of science in society. Students should be provided with opportunities to engage with all Working Scientifically skills throughout the course.

Content

Incidents, Events and Science

Inquiry question: How do science-related events affect society's view of science?

Students:

- investigate case studies of past events to consider how they have affected the public image of science, including but not limited to:
  - meltdowns of nuclear reactors
  - development of the smallpox vaccine
  - development of flight
  - positive and negative aspects of damming rivers

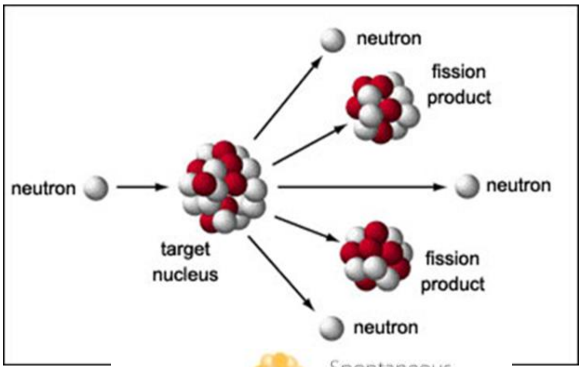
Incidents, events and science

- Society has always turned to science for the answers whenever health and safety is at risk and benefited the way societies function
- But it loses confidence in science when science is the cause of the natural disaster.

Nuclear reactions

- Generate energy through a process called nuclear **fission or fusion**
- **Nuclear fission** is the splitting of a large atomic nucleus into smaller nuclei, using neutrons, releasing a vast amount of energy in the process, mostly in the form of kinetic energy of the neutrons and daughter nuclides
- **Uncontrolled Chain Reaction**:

1. One neutron is fired into "fissile" (able to undergo nuclear fission) nucleus, such as U-235, produces the highly unstable U-236
2. Splits it into 2 smaller "daughter" nuclei: Ba-141 and Kr-92 and 3 neutrons
3. Collide with other U-235 to cause further fission events, causing an uncontrolled chain reaction to occur releasing a vast amount of energy in a very short (an explosion)



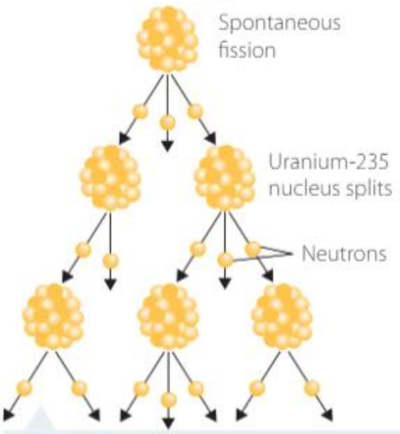
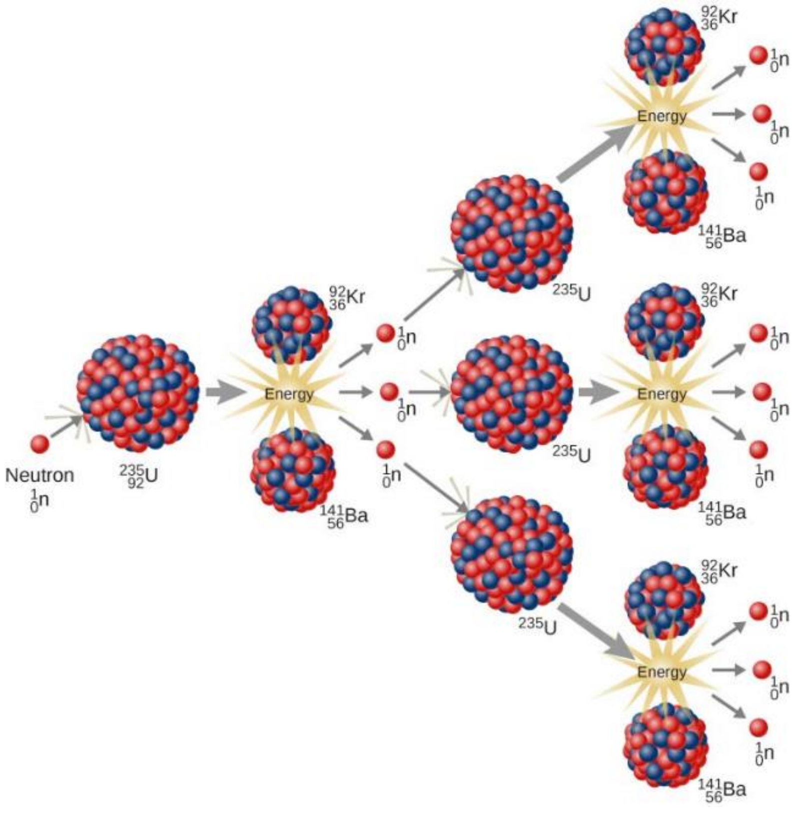
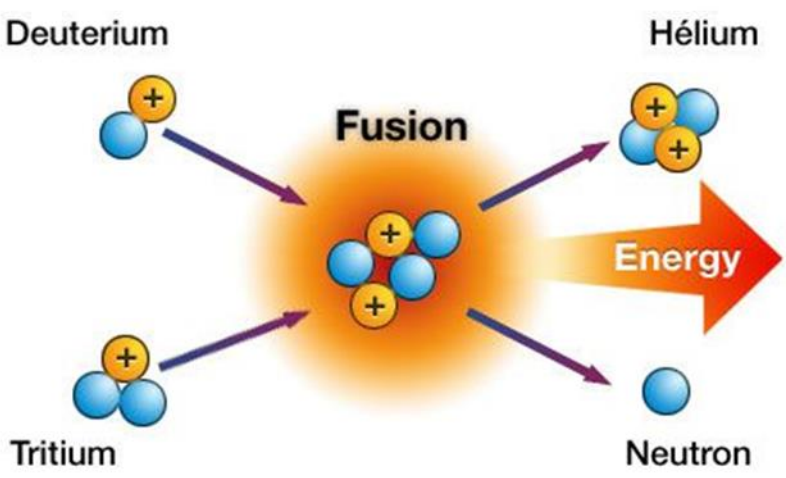


FIGURE 6.1.1 Nuclear fission is an alternative process used to generate electricity through the splitting of uranium atoms.



- A controlled chain reaction is used in nuclear reactors to produce useful heat energy
- A controlled chain reaction is where only one of the neutrons produced in each fission goes to instigate one further fission event.

- **Nuclear fusion** is the process whereby two lighter (smaller) nuclei join together to form a heavier, larger and more stable nucleus.
- Naturally occurs in the core and outer layers of larger stars because of the high temperatures and pressures required to drive the fusion process to exist there
- Energy sources of all stars that 'burn' fuel



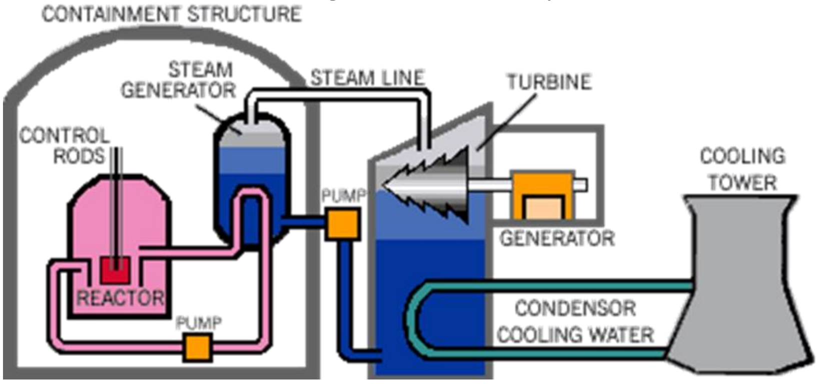
Nuclear Reactions: Fission and Fusion

- In both cases the difference in the masses of the reactants and products is the cause of vast of amount of energy production.
- This mass difference gets converted into energy in accordance with the Einstein's equation.

Nuclear reactions

- Generate energy through controlled nuclear fission of uranium atoms.
- Mass of fissile material usually U-235 or Pu – 239 is needed to produce the minimum number of neutrons required for a self-sustaining reaction
- **Fuel rods** are usually made of 3-5% enriched U-235 (embedded in uranium dioxide pellets) which are encased in zirconium allow (transparent to neutrons)
- On average 2.5 neutrons are produced per fission reaction.
    - This amount is needed as some neutrons would be absorbed by surrounding matter or escape the system and some neutrons would have insufficient energy to instigate fission even if they did collide with a nucleus
    - Too few neutrons and the reaction stops
    - Too many and the reaction rate could quickly escalate exponentially in an uncontrolled way. Nuclear reactor can explode or undergo nuclear meltdown too much heat melting core
- Moderators, such as graphite, water or heavy water, are used to slow the neutrons down.
    - These slowed or 'thermal' neutrons spend more time in the vicinity of a target nucleus and hence more efficient at instigating fission.
    - In a reactor core the moderator surrounds the fuel rods.
- **Control rods**
    - Excess neutrons are absorbed by neutron-absorbing material such as cadmium and boron, encased in steel.
    - Inserted into the reactor pile to slow down the reaction or removed to allow more fission reactions to occur as the fuel tods become more depleted

    - An emergency set of control rods are suspended above the reactor that can fall under gravity, into the core and stop the chain reaction if required.
- Coolant (water, heavy water, air, helium or liquid sodium) is used to extract the heat from the core which is then used to drive turbines to generate electricity
    - The reactor coolant undergoes heat exchange with a second cooling system to minimise the possibility of radiation leakage from the reactor
- Containment Systems
    - Powerplants use a multiple fission product barrier system to deal with containment issues. Barriers close off the fuel cladding and the reactor vessel
    - The final barrier of the reactor is made of thick concrete walls to stop highly penetrating radiation such as neutrons and gamma-rays from escaping the building
- Fission produces heat which is used to produce steam.
- Steam turns the turbine to generate electricity



- Big advantage of nuclear reactors – absence of greenhouse gas emission
- Issues: What if something went wrong?????????
  Scientists were able to sell the idea because of its advantages but couldn't foresee the implications that would occur if anything went wrong.

Chernobyl Meltdown (1986)

- Largest uncontrolled release of radioactive material ever recorded.
- RBMK 1000 reactor was undergoing testing to determine how long turbines would spin and supply power to the main circulating pumps following a loss of main electrical power supply before the routine shut down.
- During the test, the operators with inadequate training disabled the automatics shutdown mechanism.
- When the reactor became unstable due to a design flaw in the reactor and worker's failure to follow safety regulations, there was not enough time to disable it manually.
- Increased pressure blew off a 1000-ton lid on the reactor causing radiation to leak and sending air inside causing fire as the graphite-tipped control rods ignited
- A second explosion happened when hydrogen from hot water and steam contacted zirconium from the fuel rods killing two workers and released further radioactive material in the atmosphere.
- The reactor continued to release toxic material for 10 days.
- 28 people died within 3 weeks and many further deaths later.

Main reasons for the accident:

- Defective reactor design
- Insufficient staff training and communication
- Failure to follow safety regulations

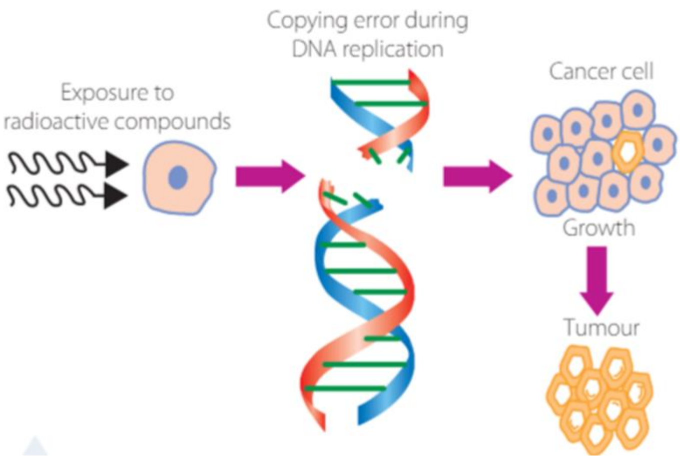
Implications to society

Immediate:

- large quantities of radioactive substances were released into the air for about 10 days. This caused serious social and economic disruption for large populations in Belarus, Russia, and Ukraine.
- Radiation doses on the first day caused 28 deaths
- Exclusion zone was set up across three heavily contaminated towns within 30km of the nuclear plant site and over a million people had to be relocated.
- During clean up over 80 workers and other personnel contracted acute radiation syndrome (ARS) due to exposure to radiation and died. This put further pressure on the healthcare system from the hospitalisation of people with ARS.
- Loss of income for power plant workers and others who earned a living through agriculture in surrounding areas

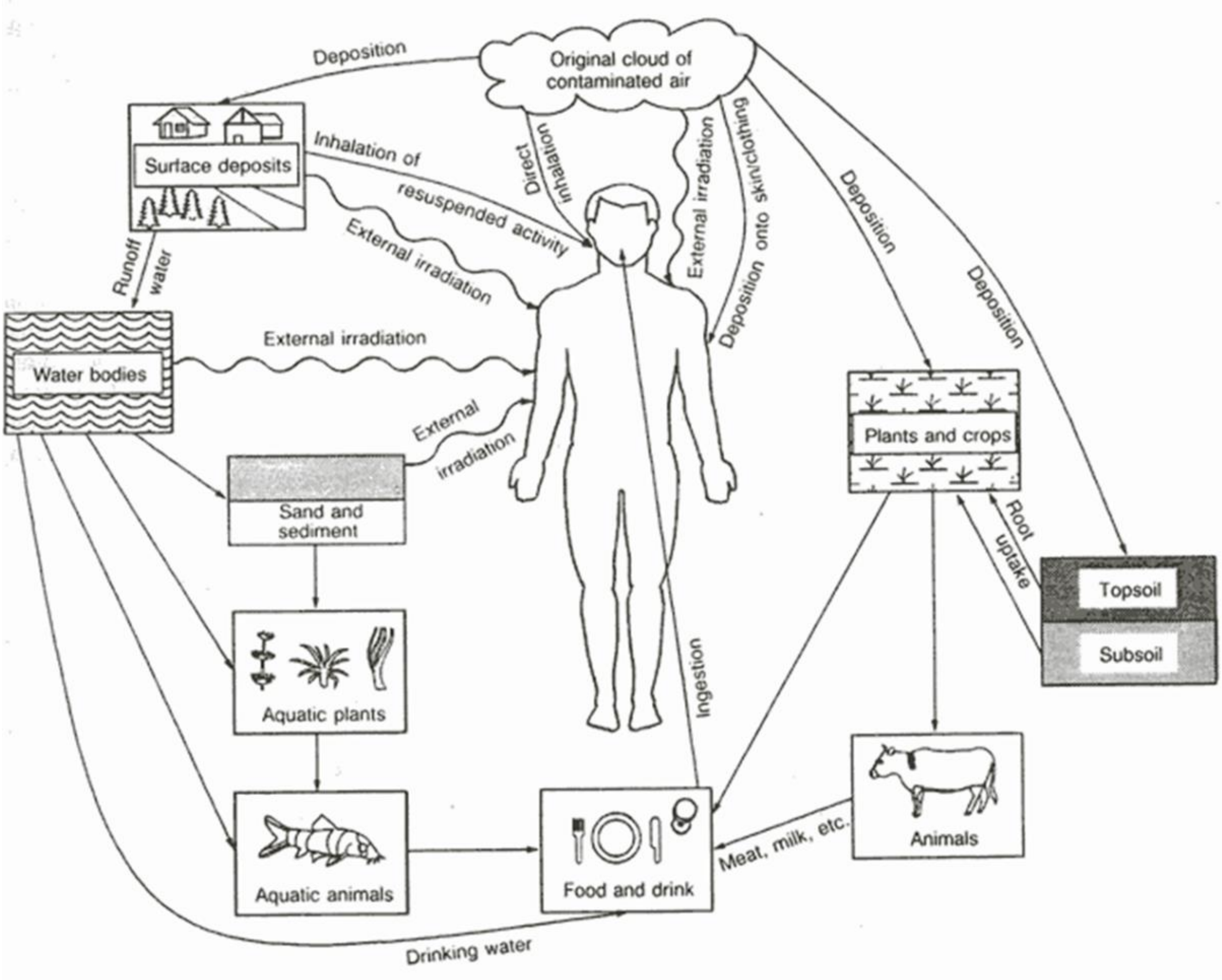
Long term:

- Accumulation of radioactive compounds such as plutonium, iodine and strontium in
- Most of the released material was deposited close by as dust and debris, but the lighter material was carried by wind over Ukraine, Belarus, Russia, and to some extent over Scandinavia and Europe. Once released into the environment, people were exposed to radioactive products in a few ways.
- Inhalation and irradiation of suspended particles were a pathway for human exposure to nuclear products. These radioactive compounds also accumulated environment and in foodstuffs, which were subsequently ingested.
- Within 15 years:
    - 4000 cases of thyroid cancer in children due to over exposure to radioactive iodine.
    - Iodine deposited over pastures where cows grazed over the land. They consumed that iodine that got transferred into milk which was bottled and distributed.
    - Early detection helped successful treatment in most of the cases but it imposed a lot of pressure on the healthcare system.



The proximity of the zone to the nuclear reactor caused the ecosystems in the zone to be **free from human interruption** since it was deemed unsafe for human habitation. The original wetlands were largely drained for agriculture by a series of canals in the mid 20th century. But since the disaster, beavers had returned to the area and started damming the canals, restoring the wetlands. This allowed new flora and fauna such and eagles and storks to flourish.

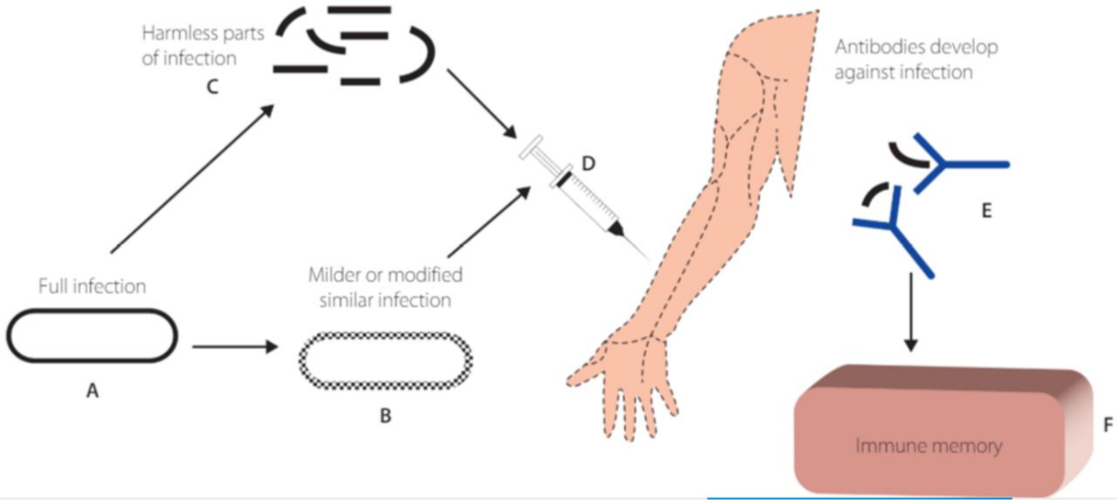
In the abandoned areas, many species succumbed to radiation and experienced detrimental genetic changes and reproductive issues.
Over 350 animals on the surrounding farms have serious deformities
However, the most severely affected organisms die before they are able to reproduce meaning that unfavourable genetic mutations are gradually decreasing. Although species diversity initially declined, the appearance of some species such as wolves that have not existed in the area for a long time started to appear.
Radioactive elements in the ground bioaccumulate from decomposers to apex predators e.g. radioactive elements -> fungi -> wild boars -> wolves

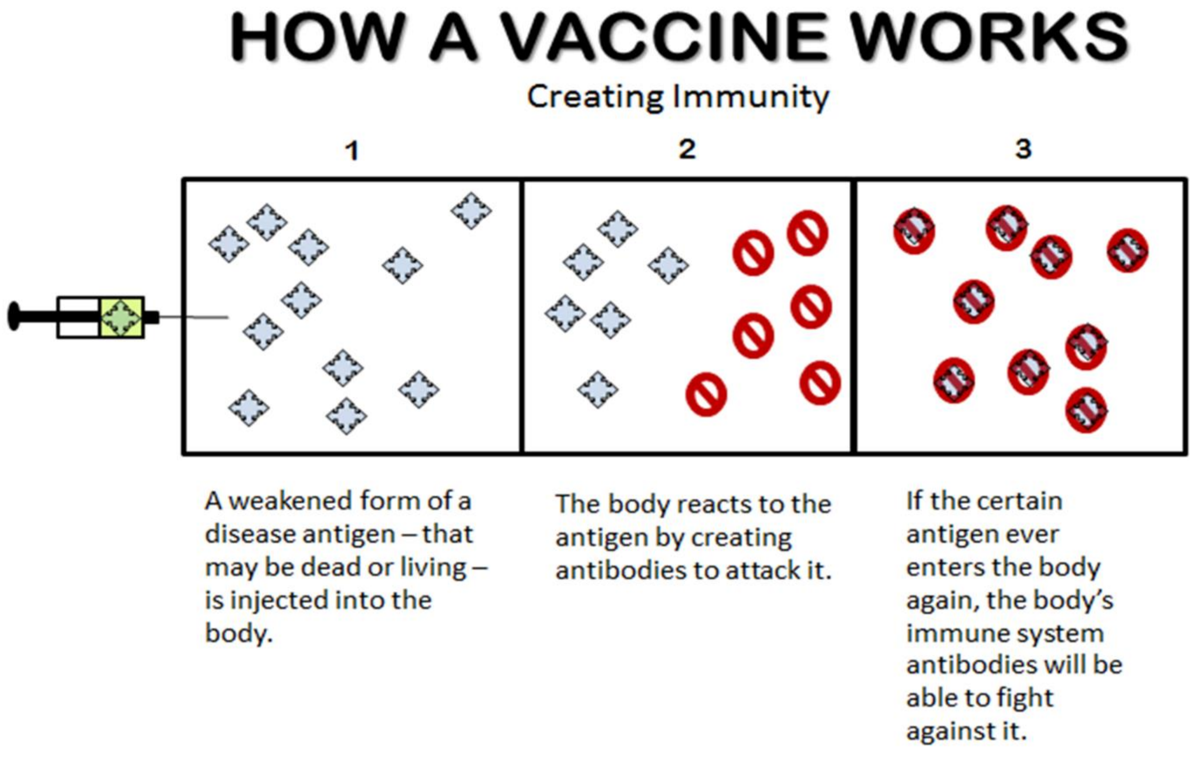


Main environmental pathways of human radiation exposure
[Source : IAEA technical report ISBN 92-0-129191-4 Vienna 1991]

Vaccination

<u>Vaccine</u>: - A weakened form of an antigen (harmless or the modified mild part of the infection) or a synthetic substitute that initiates an immune response to stimulate the production of antibodies and memory cells in the immune system. This allows the body to fight the harmful infection more effectively.





<u>Vaccination: a topic of social debate</u>

Vaccination: -

- The process to develop immunity against specific disease through inoculation with vaccine.
- Saves millions of lives every years
- Results of collaboration between scientists and many years of trial and error

Despite the scientific evidence, this technique is considered controversial and divides the opinions

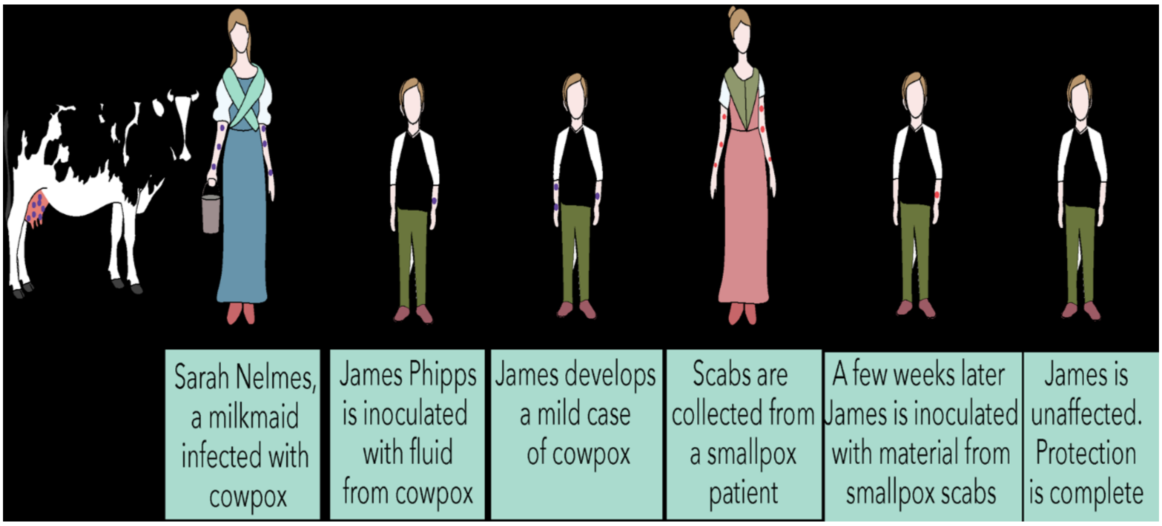
- Controversial technique of introducing weakened strains of an infection through a serum injected intramuscularly

The first vaccine - Smallpox:

- caused by virus
- Incubation period: 1-2 weeks, period between initial exposure to an infection and appearance of symptoms
- Symptoms: fever, vomiting, nausea, abdominal cramps, severe back ache, chills, headache
- As fever goes down, a rash appears on the face and spreads to the rest of body which develops into blisters filled with pus that leak the deadly, contagious virus
- The deaths from smallpox involved destruction of liver and spleen.
- As there was no effective treatment before vaccine, it became an epidemic which generated fear in society and great pressure on the healthcare system
- Prior to vaccination treatment attempts were made through herbal remedies and 'cold treatment' where rooms were left with windows open throughout the night and patients were advised not to wear clothes above their waist

Breakthrough: via simple observation

- Edward Jenner observed that milkmaids exposed to similar condition in cows (cowpox) did not appear to have any symptoms of smallpox.
- Jenner designed his experiment where he extracted pus from a cowpox blister and inserted it into an incision on the arm of eight-year-old James. Later he exposed James to smallpox (highly controversial!!!!!!)
- Though smallpox was particularly contagious to young children, but no symptoms were observed in this case



<u>However, Royal Society UK rejected his findings due to lack of evidence</u>

Jenner:

- Continued his experiment on many children including his own 11-month-old son.
- Published his findings 3 years after his first experiment.
- Coined the term 'vaccine' from Latin word vacca for cow

<u>Still it took more than 200 years and hundreds of millions of deaths before vaccine was accepted and smallpox was eradicated.</u>

- Jenner's experiment was considered 'REPULSIVE'- injecting material from diseased animal into healthy human being
- Other doctors across America and Europe adopted his technique and observed great decline in smallpox cases and gave hope for future world health

In early 1900s, two major outbreaks occurred as two stronger strains of the disease emerged

- In 1958, WHO launched a $2.4 million immunisation campaign to eradicate it globally.
- It proved difficult to achieve in developing countries due to lack of resources and insufficient supplies
- It took longer than expected but eventually it was declared eradicated by WHO in 1980

Current Research

- WHO and scientists wanted to continue research on the virus.
- Virus is held at two locations- the US Centers for Disease Control and Prevention in Atlanta and the State Research Centre of Virology and Biotechnology in Russia

Justification:

- It allows scientists to find out more about the development of the virus and trace its genetic history – found out to be 8000 years old
- Concerns: not all health officials consider having alive virus as ethical and debate the possibility of accidental release or use of stocks as biological weapons.

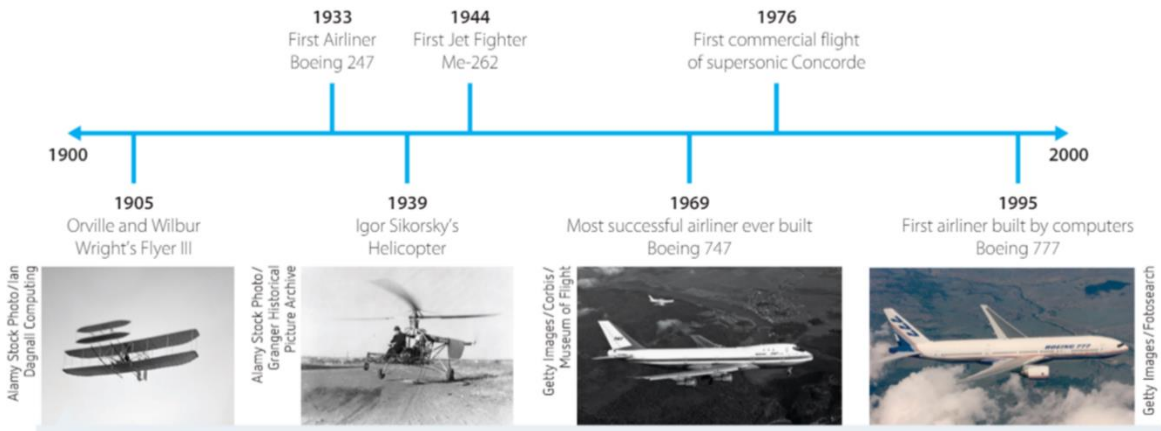
Implication of eradication:

- Led the way for eradication of polio
- Reduction in cases of whooping cough, measles, mumps, rubella, hepatitis B, diphtheria, tetanus and pneumococcal infections
- Helps society to stay healthy
- Reduces strain on healthcare system, hospitals
- Saves taxpayers money on treating people.

Development of Flight

Benefits:

- Reduces global distance for
  - Face to face meetings
  - Improvement of communications
  - Facilitates executive meetings in international corporations
  - Allows employees to attend professional development opportunities and conferences
- Coordinates developed society



1000 BCE – First Kites are invented in China.

1485–1500 – Leonardo da Vinci designs flying machines. He made many drawings of wings, parachute flying machines.

1709 – Bartolomeu Laurenço de Gusmao designs a model glider that could carry people over long distances. It was based on bird anatomy and referenced by the wright brothers

1783 – The first untethered manned hot air balloon flight was on 21 November 1783, in Paris, France in a balloon created by the Montgolfier brothers.

1843 – George Cayley's biplane design is published.

1895 – Otto Lilienthal flies biplane gliders.

1903 – Orville and Wilbur Wright make the first recorded powered, sustained and controlled flight

1927 – Charles A. Lindbergh completes first solo, nonstop trans-Atlantic flight.

1930 – British inventor Frank Whittle invents the jet engine.

1933 – A modern airliner, Boeing 247, flies for the first time.

1939 – Germany's Heinkel 178 is the first fully jet-propelled aircraft to fly

1947 – Charles E. Yeager pilots Bell X-1—the first aircraft to exceed the speed of sound in level flight.

1957 – Soviet Union launches first man-made earth satellite, Sputnik 1.

1961 – Soviet cosmonaut, Yuri Gagarin, is the first man in space

1969 – U.S. astronauts Neil A. Armstrong and Edwin E. Aldrin, Jr., are the first to walk on the moon

1970 – Boeing 747 makes the first commercial flight. Most successful airliner ever built

1976 – Concorde begins its first passenger-carrying service.

1995 – Boeing 777 first aircraft produced through computer-aided design and engineering

2007 – Airbus release the A380: a double decker civilian passenger jet

Malaysia Airlines Flight 370 was a scheduled international passenger flight operated by Malaysia Airlines that disappeared on 8 March 2014 while flying from Kuala Lumpur International Airport to its destination, Beijing Capital International Airport.
Date: 8 March 2014
Summary: Cause unknown, some debris found Passenger count: 227

2016 – Solar Impulse 2 is the first plane powered by a renewable energy source to tour the globe

March 10, 2019: Ethiopian Airlines Flight 302, a 737 MAX 8, registration ET-AVJ, on a flight from Addis Ababa, Ethiopia to Nairobi, Kenya, crashed 6 minutes after take-off, killing all 157 people aboard: 149 passengers and 8 crew members. The plane was only 4 months old at the time of the accident. In response, numerous aviation authorities around the world grounded the 737 MAX, and many airlines followed suit on a voluntary basis. Caused by MCAS manoeuvring characteristics augmentation system malfunction in the software of the airplane caused it to nosedive

River Damming

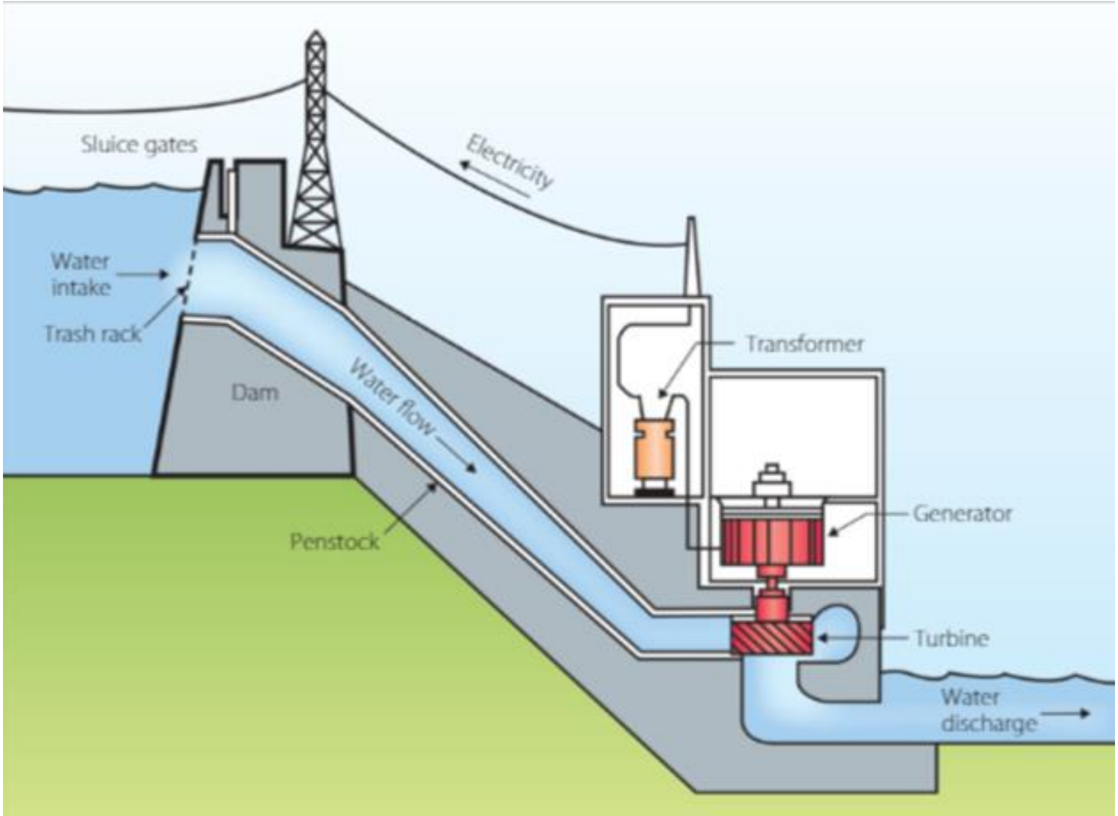
Damming: the process of blocking a river using a large wall made of rock or concrete e.g. Warragamba

Benefits:

- Dams create water reservoirs to supply water required for agriculture, irrigation, domestic use and public recreation- boating, water skiing etc
- Alternate power source- hydro-electricity- generates clean, renewable energy in controlled manner without greenhouse gas emission

Hydroelectricity:

- when demand is high, gates are opened and water flows with higher energy from the higher reservoir on the turbine connected to generator and transformer to the lower reservoir.
- When enough power has been generated, gates are closed, and water is pumped back to the higher reservoir.
- In 2016, it contributed 42% in Australia's and 71% to world's renewable energy.



Limitations of damming rivers:

- Changes free flow of water
  - Changes physical properties of rivers- amount of dissolved oxygen, temperature, chemical composition, salinity etc
  - organisms may not be able to cope with the change in their ecosystem- leads to loss of biodiversity
- Disconnects different water ways:
  - Prevents fish migration
  - Overpopulation of aquatic animals that are lower in food chain
  - Loss of biodiversity
- Blocks waterways:
  - Stops natural sedimentation downstream
  - Prevents natural maintenance of deltas, wetlands

Regulation of Scientific Research

Inquiry question: Why is scientific research regulated?

Students:

- investigate the need for the regulation of scientific research in, for example:
  - genetic modification of sex cells and embryos
  - development of biotechnological weaponry
  - testing of pharmaceuticals
  - products and processes of the nuclear industry
  - protection of Indigenous cultural and intellectual property
- investigate and assess ethical issues surrounding current scientific research in, for example:
  - use of radiation
  - pharmaceutical research
  - gene manipulation in biotechnology
  - mining practices
  - bioprospecting
- investigate a range of international scientific codes of conduct in regard to scientific research and practice in the areas of, for example:
  - cloning
  - stem cell research
  - surrogacy
  - genetically modified foods
  - transplantation of organs
- evaluate the effectiveness of international regulation in scientific research and practice

Regulating Science

Regulation ensures that all scientific research and activities are:

- Carried out ethically
- Governed by laws for safety and human rights

Morals
- Internal principles
- Personal beliefs or judgement of ‘right/wrong’

Ethics
- External rules
- Concepts of ‘right/wrong’ developed by society and governed by legal guidelines

- Ethics arise from interaction between science and culture. Hence, they differ between cultures. It poses problems for international code of conduct.
- Hence overarching association develops the guidelines to be followed by all the countries which can be adapted to form their own laws and regulations.

Genetic Modification:

The process of manually adding new DNA to an organism.

identification	Extraction	cloning	Modification	Transformation	Breeding
• Organism with desired trait is identified.	•DNA is extracted	• Desired gene is located and copied (cloned)	• Cloned DNA might be modified to make it more efficient	• Injecting the modified DNA (transgene) into the cells of the recipient via a process called transformation	• For successful transformation, the new transgenic organism is bred

Genetic modification of sex cells and embryos

- Genetic testing is vital for the success of IVF where embryo is tested before implantation to reduce the chance of gene disorders such as heredity breast or ovarian cancer.
- Science is taking it further with modification of sex cells and embryos
  In 2017, scientists in China reported that they have modified a human embryo to remove a mutant strand of DNA that causes beta- thalassemia and replaced it with functional DNA. Beta- Thalassemia is inherited, life threatening blood disorder that destroys red blood cells.
- Still experimental stages – scientists reported many copying errors that could lead to unpredicted complications in viable embryos.

Pros: prospects of eliminating hereditary diseases

Cons: technology could be abused and can become a way for parents to design their own offspring including choice of sex and other characteristics, such as eye and hair colour

The technology is strictly regulated to ensure it doesn't become gateway to make designer babies.

Australia's National Health and Medical Research Council:

- Prohibits human genetic cloning and associated technologies.
- Enforces strong penalties for non-compliance including large fines and possible ban for many years from carrying out further research

Gene therapy:

genetic manipulation in biotechnology uses genes to target areas of the body affected by disease.

Cancer treatment:

- Cancer is the rapid and uncontrolled proliferation of cells. Metastatic – invade other tissues
- These cells are identified as cancerous when the genetic material responsible for regulating the cell division is mutated
- Research is being carried out if the mutated genetic material can be replaced with transgenes that can produce cancer killing proteins.

Severe combined immunodeficiency SCID:

- A disorder that disrupts functioning of vital Lymphocytes (white blood cells)
- People with SCID are very susceptible to infection and must live in a sterile environment.
- Scientists successfully treated this disorder with gene therapy by inserting a functional variant of the gene into a viral vector which transforms extracted bone marrow cell that belonged to the patient. Returning these bone marrows cells back to the patient allowed their immune system to produce functional lymphocytes

Ethics in genetic manipulation:

- Experimentation leading to the attempt to reduce the chance of congenital diseases
- E.g. introduction of germline gene therapy: the process where DNA is transferred to the cells that produce reproductive cells
- Concerns:
    - May cause unforeseen or irreparable effects to the developing foetus
    - it is also violation of basic human rights and foetus does not have the ability to accept or reject the therapy.

Bioprospecting:

- conducting tests to see if a particular plant or animal could be used to make medicine or other products.
- could use entire organism or a gene/compound extracted from that organism
- Economic and medical benefits are clear, and process has been successful in the past.
- However, there is risk of exploitation and extinction of species that are being used for medicine.

Ethical concerns over:

- Removal of an organism from its natural habitat and its genes in the name of testing
- Ownership of natural resources

If an organism is found to be of medicinal value, then:

- Do corporate representatives have the right to raid entire ecosystem of this resource?
- How much of the resource can be taken from an ecosystem before it becomes detrimental to the environment?
- What happens if the resource is found in polar regions which are controlled by multiple stakeholders?

Interrelationship between stakeholders of a useful biological resource

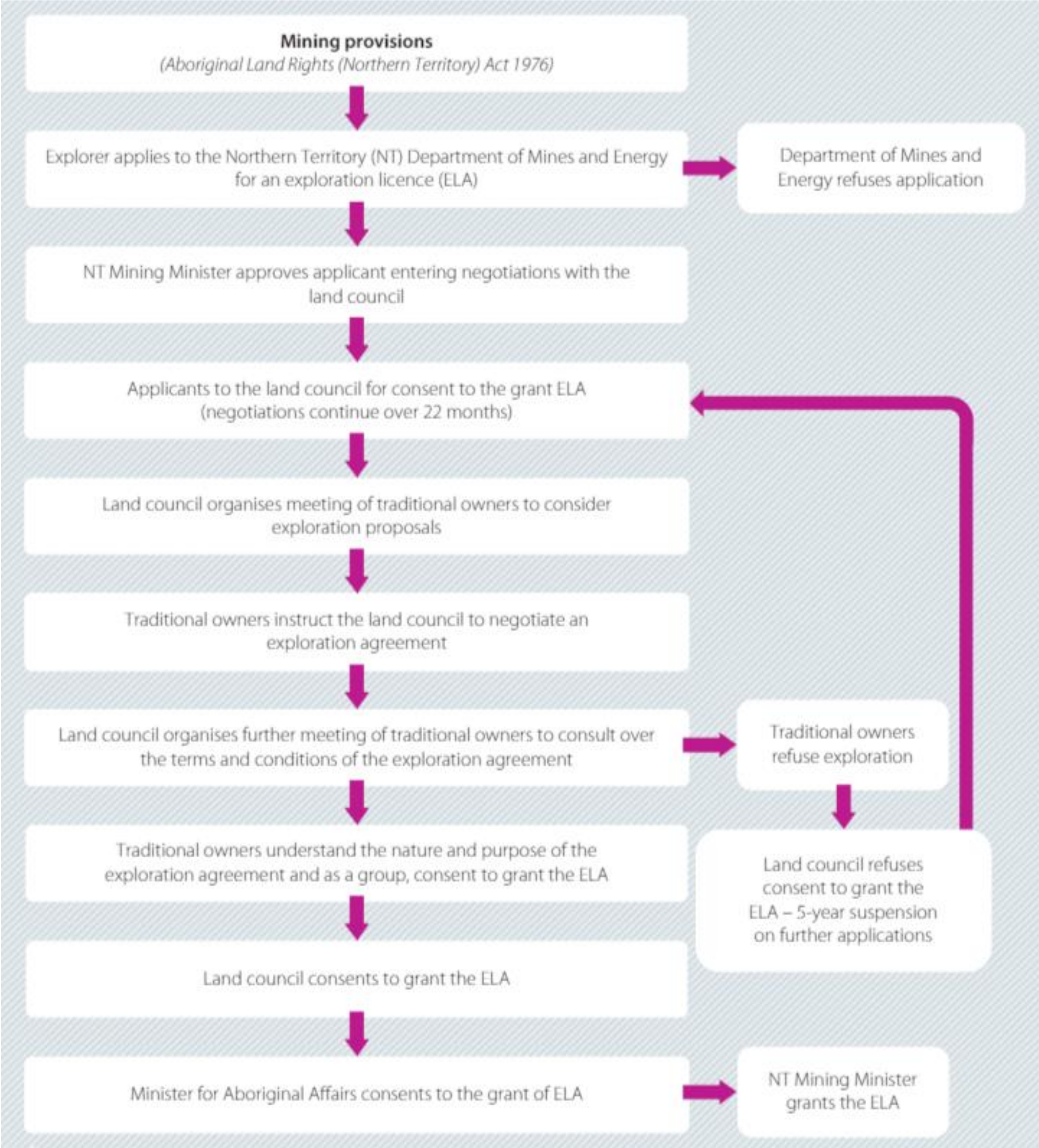
Protection of Indigenous culture and intellectual property (ICIP)

- Indigenous people are aware of the assets of their environment and have already been applying the knowledge for thousands of years.
- Taking natural products from their land for commercial purposes raise many ethical issues over ownership of intellectual and cultural property.
- Regulations have been established to protect ICIP from exploitation by big businesses and this property legally remains under the ownership of traditional custodians of the land including artwork, folklore, vital medicinal plants, animals and minerals, etc.

Local communities oppose the idea of:

- Exploiting the land that has historical value
- Disrupting the resting place of their ancestors

Mining on Aboriginal land- all permits have to be approved by traditional owners of the land



Scientific code of conduct

Set of rules and regulations that allows scientists to work without crossing major ethical boundaries

Field of biotechnology:

- Manipulates natural processes as it uses cellar processes to develop products to enhance and prolong our lives
- Medical Technology Association of Australia oversees to ensure our research complies with international standards.

Stem cell research:

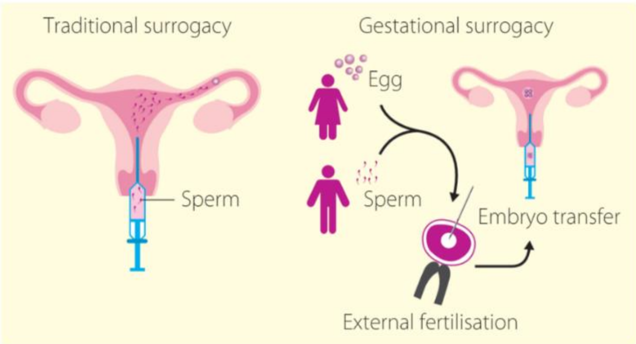
- Cells carry out a particular specialised function and become specialised during in-utero development and once thy have been differentiated they are unable to reverse.
- Stem cells are cells that have not yet differentiated in the 4-5-day old embryo but can develop into many different types of cells e.g. skin, blood bone
- Differentiated tissues are cultured in petri dishes to test the effects of drugs and other products on human tissue gaining valuable information with no adverse effect on any human being
- **Ethical Issues:**
  - removing stem cells from the embryo destroys it and its potential to develop into human being.
  - Embryo cannot accept or reject the testing- basic rights are denied
- Hence stem cell research needs strict regulation and code of conduct that includes ethical, efficacy and safety principles that must be followed. Through the International Society of Stem Cell Research sets the current international rules regulations and code of conduct for stem cell research. Includes the ethical, efficacy and safety principles.

Surrogacy:

Reproductive technology where the woman that caries the embryo and gives birth is not the intended parent.

Traditional surrogacy: women who caries the baby uses her own egg and sperm form the father to make the embryo

Gestational surrogacy: an egg from the biological mother fertilised by sperm form the biological father using IVF and that fertilised embryo is implanted into a different woman



Sought by parents for many reasons:

- Women who are unable to carry a baby to full term
- Older women
- Women with medical issues
- Same sex couples

Commonwealth government published document outlining the laws and legislations of surrogacy in Australia called the 'Surrogacy Matters: Inquiry into the regulatory and legislative aspects of international and domestic surrogacy arrangements'

It includes Australia's obligation to offshore treaties including the convention on the rights of children as well as on anti-trafficking and organised crime treaties.

Organ transplantation:

- Process involving removing an organ from one person (usually deceased) and transplanting into another individual.
- To be placed on transplant wait list, all other treatment options must have been exhausted and medical assessment should have been undertaken to confirm that their own organ is failing
- Most common organs needed for transplantation are heart, lungs, kidneys, live, pancreas…
- Allocation of transplant in Australia does not discriminate based on race, religion, gender, disability, social status, age or economic status.

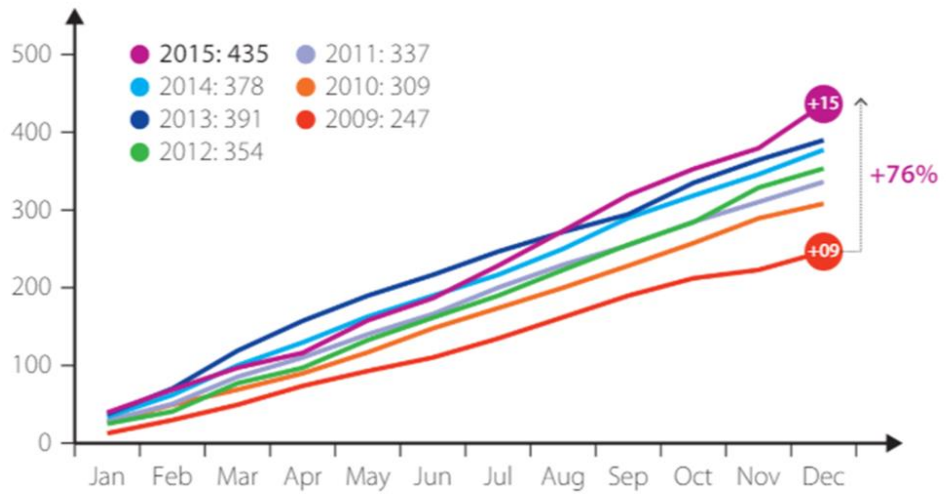
Concerns: Financial and societal pressure

- Organ shortage in many countries as people believe that body is sacred and must be cremated/buried intact.
- For Living donor- recovery time can be up to 6 weeks. If donor is not entitled or exhausted their paid leave this can lead to 6 weeks without pay
- 'Supporting leave for living donors program' - 2 year pilot program implemented in Australia in 2013 – to reimburse employers for any leave payments given to employees who have taken leave for live organ donation procedure
- Program extended for further 2 years (2015-17) and paid leave has been increased from 6 weeks to 9 weeks at National Minimum Wage.

Concerns: Black Market Trade

- Small number of available organs has led to black market trade exploitation as:
    - Organ trafficking
    - Poor people selling their own organs (recipients of organs paid up to $20000 but donor received less than 5%)
- There is no international body governing organ transplantation and individual countries have their own regulations and policies to reduce ethical procedures
- WHO approximated 10% of total kidney transplants worldwide were obtained and performed through the black market
- 'DonateLife' Australian initiative has shown 76% increase over 6-year period in individuals electing to donate organs after their death
- "Ethical guidelines for organ transplantation from deceased donors"- document published by NHMRC in 2016 highlights:
    - Ethical practice
    - Human rights
    - Criteria for eligibility, allocation and monitoring of practice.

Since initiation of 'DonateLife' program <u>Deceased organ donors 2009-2015</u>



- In 2015 there were 435 deceased organ donors, the highest donation outcome achieved in Australia.
- Represents a 15% increase over 2014 (378) and a 76% increase over 2009 (247), the year the DonateLife Network was established.
- When compared to the historical average of 205 deceased organ donors per annum (2000-2008), the 2015 outcome of 435 donors represents a 112% increase.

Emerging technologies
To combat black market:

- Organ creation using 3D printers – current prototypes for ears and nose
- Research for more complex organ creation is ongoing

Facial Transplants:

- new medical innovation- to give normal appearance to victims of burns, trauma or genetic malfunctions
- ethical issues and emotional issues – people seeing a dead family member or friend's face

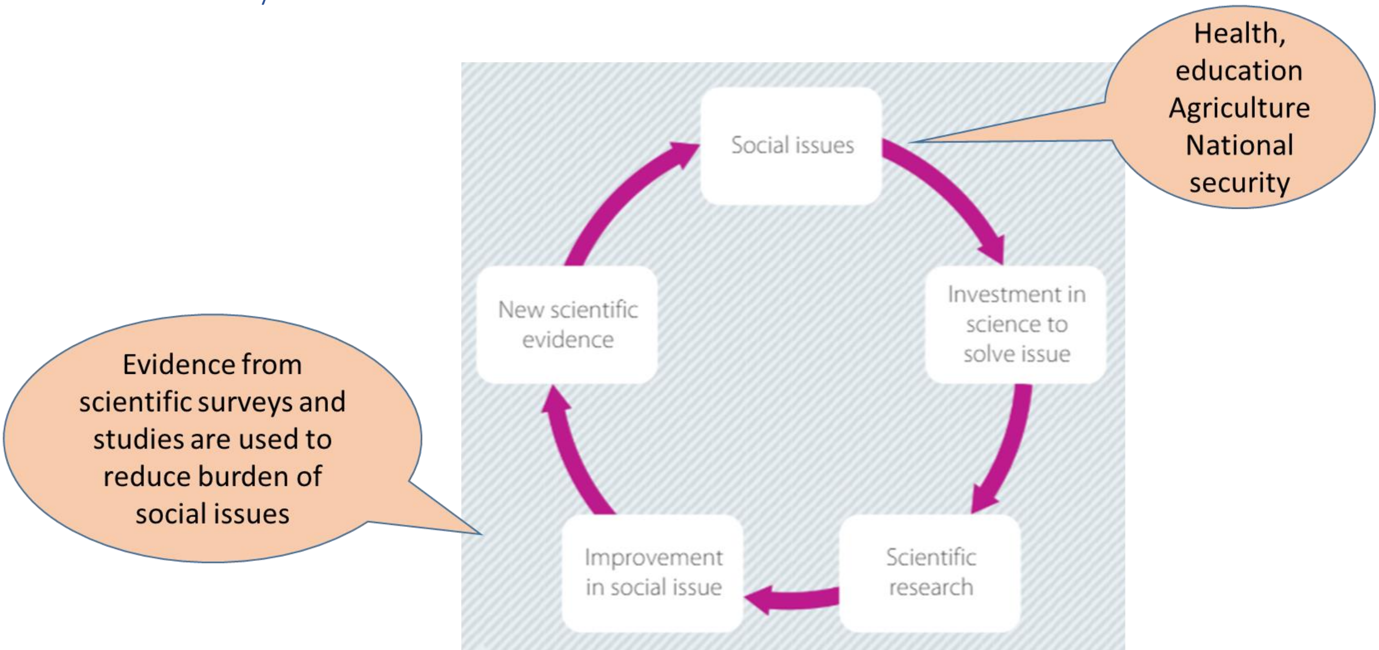
Influence of Economic, Social and Political Forces on Scientific Research

Inquiry question: How do economic, social and political influences affect scientific research?

Students:

- evaluate the costs involved in space exploration compared to investments in social issues, for example poverty and human global food supply
- evaluate how scientific research aids economic development and human progress in relation to, for example:
  - nuclear power generation
  - use of antimicrobial drugs
  - genetically modified foods
  - use of petroleum products
  - robotics and the use of drones
- evaluate the impacts of scientific research, devices and applications on world health and human wellbeing, including but not limited to:
  - medical surgical devices
  - surgical procedures
  - water purification and wastewater treatment
  - vaccination programs for the eradication of disease
- using examples, analyse the impacts that governments and large corporations have on scientific research, including but not limited to:
  - corporations and market opportunities
  - university research project budgets
  - governmental budgets and limited time priorities
  - benefit-sharing in research using Indigenous intellectual and cultural property
- evaluate how personal, cultural and socioeconomic perspectives can influence the direction of scientific research, for example:
  - perceptions about diet in a multicultural society
  - investigating traditional medical treatments
  - mining practices

Science and society



- Research is a response to improve social issues which in turn lead to further issues and a need for more research
- Scientific research is costly and rely on more funds from universities, the government and grants from corporate bodies that have an interest in the concept being researched as well as scientific organisations themselves
- Relying on external agencies for funding raises ethical concerns over the possibility for bias or skewed representation in published data

Controversial investments – space exploration

- What does society get from space exploration?
- How does society benefit from money spent on it?
- E.g. in 2016- US: $6 billion to NASA – to determine the presence of water source on Mars that can be accessed thousands of years in future
- Should money be invested in fixing more pressing and current issues- hunger and poverty in developing countries

FOR	AGAINST
▪ Space exploration has the potential to solve current issues such as harvesting solar winds as an alternative energy supply ▪ Space exploration is important for the future of humankind as it may determine whether there is a possibility for humans to live elsewhere if our planet is no longer habitable ▪ Astronauts are capable of testing the fundamental principles of physics ▪ Space travel by tourists can be used to boost the economy ▪ Scientific knowledge is a valuable resource	▪ Scientists could be using their expertise to solve more pressing issues such as poverty and hunger ▪ The current needs of people on our own planet should take precedence over the potential of other planets ▪ There are so many areas of our own planet that have yet to be explored ▪ There have not been any direct benefits of space travel ▪ Space-related inventions such as equipment that functions at zero gravity are irrelevant to normal daily life ▪ Money is only invested in space exploration to 'show off' to other countries

Nasa developed memory foam in mattresses used to improve safety of aircraft cushions, freeze dried food, preserve quality extending shelf life and low temp help minimise deterioration actions retaining nutrients, LED light therapy in cancer treatment
GPS, mobile phones to connect the world

Science and economics

- Research in the areas such as nuclear power, genetically modified foods, pharmaceuticals, robotics etc have potential long term economic benefits

Genetically modified foods

- Genetic modification: involves inserting of favourable gene into the cells
- Benefits: improved yield, longer shelf life for the fresh produce, save money on herbicides
- E.g. *Flavr Savr* tomato involved silencing the genes responsible for ripening. Resulted in the tomato lasting up to 45 days without becoming overripe extending its shelf life

Potential for genetically modified food – 'Golden rice'

- Golden Rice- GM variety to prevent millions of deaths due to Vitamin A deficiency in developing countries
- New varieties can supply up to 60% of the RDI while before it only supplied 5-8%

Drones

- Used:
    - Where manned flight is too dangerous or expensive
    - Have been used to assist military
    - Storm forecasting, mapping, surveillance, law enforcement, wildlife tracking
- Being developed as a tool to assist surf lifesavers – drones fixed with cameras to identify dangers: rips, sharks. They can also carry lifesaving equipment ready to be released if needed.
- Visual information collected from drones can be analysed and merged with 3D visual tools and predictive computer technology to generate maps

Science and world health

Science can determine better ways to control and maintain the risks in health and wellbeing

Vaccination programs

- Used to control potential outbreaks for new diseases or new strains
  eg: Ebola vaccine trialled in December 2016 – deemed effective
- Eradication of diseases and maintaining that eradication e.g. smallpox
- Australian initiative for pertussis (Whooping Cough) vaccination to all pregnant women as antibodies travel across placenta as it was found that half of all cases was infected by family members and parents.
- In the trimester of pregnancy, the newborn can be protected from the sisease before receiving their 6-week vaccinations as antibodies created by the mother in response to the vaccination cross the placenta
- Recommendation – all people that would come in contact with baby in the first 6 weeks should get a booster dose

Surgical procedures and devices

Development of new surgical procedures with minimal invasion have improved wellbeing by:

- Less trauma to the body
- Reduced chance of infection
- Short hospital stay post-surgery
- Reduced pain and recovery time
- Decreased scar tissue

Keyhole surgery

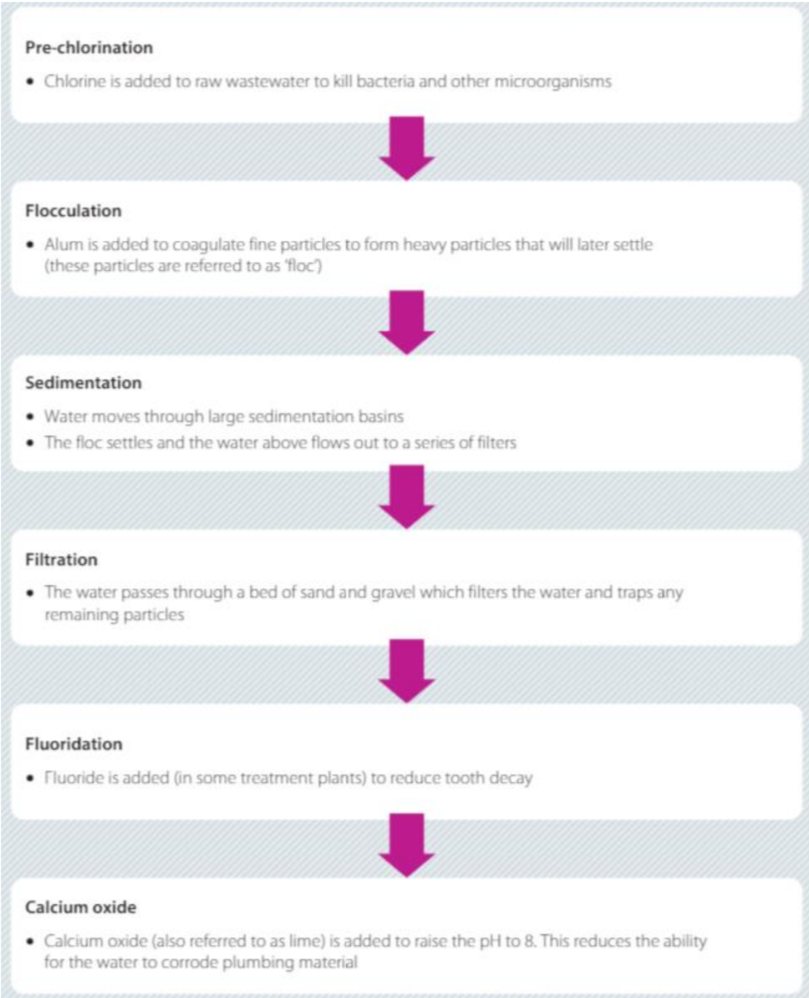
- Surgery with minimal invasion, through incision about 1 cm, developed in early 20th century
- Uses long, thin instruments fitted with light and video camera that relays the information required by the surgeon.
- Eliminates the need to open up the abdomen and therefore reduces the pain, discomfort and recovery time for the patient
- As camera technology improved so too did the quality of feedback
- Now most preferred technique and used whenever feasible

Cataract surgery

- Started with Indigenous communities in 1976, Fred Hollows was able to reduce avoidable blindness in remote areas without adequate access to healthcare.
- Fred Hollows - Australian ophthalmologist who restored the sight of elderly community members in remote areas of developing countries by performing cataract surgery in bulk
- Surgery involves replacing the damaged lens with an inexpensive device called an intraocular lens (IOL) which performs the same function as the natural lens
- Low-cost allows 20-minute surgery to be completed by a ophthalmologist for as little as $25
- Fred Hollows Foundation continues work in over 25 countries and has helped restore sight of over 2 million people worldwide

Water purification

- As population in the world increases, water must be conserved and recycled for future use
- Wastewater treatment plant makes wastewater fit to be used and ingested. Wastewater is water that has been used before e.g. domestic, commercial or agricultural settings; runoff



- Water is vital to life as it helps the body regulate temperature, protects organs and tissues, lubricates joints, removes waste products, dissolves minerals and nutrients and helps carry oxygen and allows the body to make energy

Lifestraw device

- Light and portable device – stringently tested and refined through scientific experimentation
- It allows contaminated water to be drinkable through filtration system

Political influences

- National budgets and university grants influence the amount of money available for research
- Hence there is a lot of financial pressure and competition on allocation of money and resources to specific areas of study

Marketing

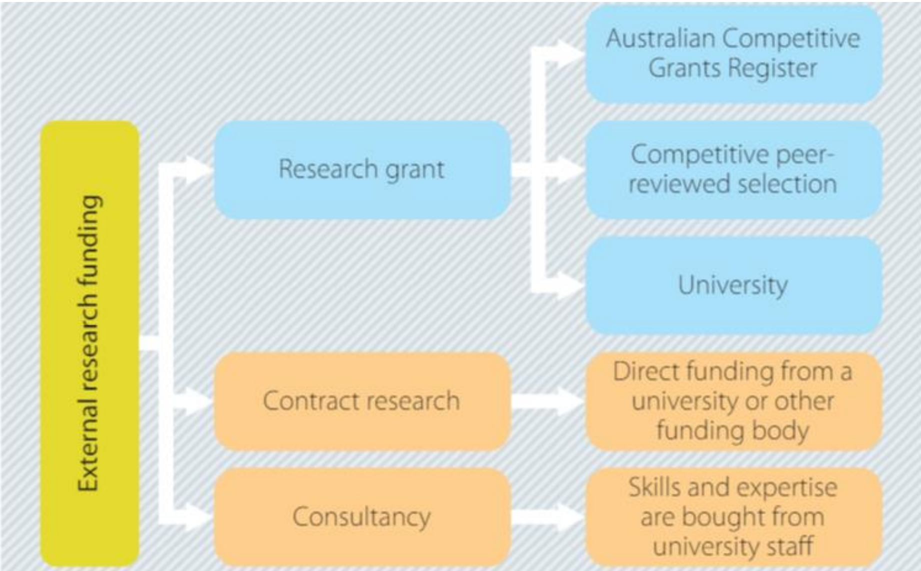
- Scientific evidence to support a product is the most influential marketing technique

Marketing in pharmaceutical industry

- Pharmaceutical industry must pass stringent regulations for research and testing.
- Due to competition, many companies choose to invest in further independent studies to get additional data to gather results that reflect favourably on their products or to highlight the necessity of the drug for the condition being studied

University research project budget

- Research depends upon budget allocated to that field of study
- Australian competitive Grants Register (ACGR) funds research that aligns with schemes funded by Australian Commonwealth Government
- Three general pathways when applying for research funding: research grants, contract research or consultancy



Government Budget

- ACGR outlines the areas of research available for government grant that are in accordance with the schemes outlined as important in the national budget.
- These areas depend on the social issues at a given time E.g. if there is financial pressure on the healthcare system from elderly patients requiring treatment and care for illnesses such as dementia and Alzheimer's, there my be more money allocated in the budget to fund studies related to these diseases
- As government releases a new budget, these research priorities will change

Benefits using Indigenous cultural and intellectual property

- By publishing ICIP, Indigenous people are sharing their knowledge with the society.
- ICIP is at a risk of exploitation if it is not treated with respect when referenced in scientific research.
- When it is planned to integrate ICIP in research, the aim, strategies and management of the project must be discussed with the indigenous community.
- A succinct, unambiguous and mutually agreed contract must be drafted that outlines research project and its potential benefits
- Contract must be free of ambiguity or information that could be misconstrued
- It must be acknowledged as a source if researchers intend to use the knowledge of Indigenous Australians

Personal, political, socioeconomic influences

- People rely on science to back up claims where there are many opinions on the same thing.

Dieting in multicultural society

- Different cultures and norms have different opinion on IDEAL BODY and diet to achieve it
- Obsession with dieting has influenced research to support or falsify the claims of specific diets designed to make a person lose fat, building muscle mass
- E.g. Detoxifying diet- cleansing diet the conclusion of peer reviewed research:
    - Current data from clinical trials is flawed with poor methodology and small sample size
    - Data was obtained through animal testing
    - No such information has been collected in human trials

Diets in history – the tapeworm diet

- 1900s England, marketed as a way to obtain ‘perfect figure’ without reducing kilojoule intake and involved eating tapeworm eggs from the jar or in the pill form.
- Once the parasite reached the intestines, it matured and could grow up to 9 metres long.
- It absorbed the nutrients and fats ingested by the host
- After reaching the goal weight, host would take a subsequent pill to remove the worm from the digestive system. it was accompanied by severe abdominal cramps.
- Research conclusion:
    - Long term ingestion of worm was found to be associated with a range of diseases- dementia, meningitis, epilepsy
    - these diseases were the result of the condition where the tapeworm larvae got to the bloodstream and lived as parasite in the brain tissue

Dieting across cultures

- Diets differ according to the dominant values of the culture e.g. areas of South pacific – Fiji, Samoa, Nauru being large was historically an attractive trait, suggesting wealth and access to resources. Recreational activities and traditional strength competitions may also encourage excess food to be taken
- Attractive female or male body is viewed differently according to a person’s culture
    - Westernised cultures accustomed to idolise slimness
    - Some other cultures view additional weight as an indication of health and fertility
- Anomalies in attractiveness between different countries can be problematic where extremely underweight or overweight individuals are favoured

Socioeconomic status and dieting

- For people with low socio-economic status, fulfilling hunger takes precedence over nutritional value
- They do not have the education or access to resources to allow them to make healthy nutritional decisions
- They generally strive on starch-based food rice and potatoes that have fewer vitamins.
- Research dedicated to nutritional and health effects of dietary staples in different countries has led to the development of foods such as golden rice to help reduce the deaths from diseases caused by vitamin deficiency.
- Journals such as Journal of Nutrition and the International Journal of Food Science do this